

**Project Proposal**  
**BSc (Hons) in Artificial Intelligence**  
**Adversarial Geometric Distillation Based Mitigating Hallucinations in 3D**  
**Model Generation**

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## 1 Introduction

The field of 3D content generation has evolved from optimization-based reconstruction to zero-shot synthesis via Neural Radiance Fields (NeRF) [1] and Score Distillation Sampling (SDS) [2]. While subsequent architectural refinements in Magic3D [3] and Fantasia3D [4] have significantly improved visual resolution and surface extractability, these models remain hindered by a fundamental lack of 3D geometric awareness inherent in 2D diffusion priors.

This deficiency leads to persistent "geometric hallucinations," most notably the Janus problem, where objects display redundant or inconsistent features [2], [5], [6]. Theoretical analysis identifies these artifacts as a manifestation of "mode collapse," where optimization settles into high-probability 2D frontal views [5]. The current state-of-the-art approach, Hallo3D [7], attempts to mitigate this by using Large Multi modal Models (LMMs) to detect structural errors and generate "enhanced negative prompts." However, a critical research gap remains - LMM driven frameworks suffer from secondary reasoning hallucinations, where the LMM misidentifies errors or fails to perceive subtle spatial inconsistencies. Furthermore, relying on a purely text-based feedback loop creates a "semantic bottleneck," as negative prompts lack the spatial precision required to guide 3D optimization effectively, often resulting in over-correction or new artifacts.

The significance of addressing these multi layered hallucinations is vital for the professional scalability of AI driven asset creation in Virtual Reality (VR), Augmented Reality (AR), and gaming, where structural integrity is as critical as visual fidelity. Current frameworks fail because they cannot ground LMM reasoning into precise 3D spatial constraints.

This research project introduces a closed loop geometric grounding framework designed to overcome the reasoning limitations and secondary hallucinations identified in Hallo3D [7]. The proposed technology moves beyond simple textual prompts by integrating LMM detection with a direct, spatially aware geometric feedback mechanism. By translating multi-modal observations into explicit structural constraints, this research aims to eliminate both primary 2D prior hallucinations and secondary LMM reasoning errors, providing a robust methodological solution for consistent and structurally sound 3D content generation within the AI literature.

## 2 Objectives

- Critical review of the literature on geometric hallucinations in generative 3D models, analyzing how the evolution of these models led to specific topological errors and structural inconsistencies.
- In-depth study of adversarial neural networks and self-distillation technologies used for automated artifact detection and mesh refinement.
- Design and develop an Adversarial Geometric Distillation system that integrates an auxiliary AI critic to guide the correction of 3D geometric anomalies.
- Evaluate the proposed solution using quantitative geometric metrics (e.g., surface smoothness, manifold quality) and qualitative performance comparisons against baseline models.

### 3 Literature Review

#### 3.1 Foundations of Neural 3D Representations and Generative Priors

The paradigm of 3D content generation has evolved from optimization-based reconstruction to zero-shot generative synthesis. The technical foundation was established by Neural Radiance Fields (NeRF) [1], which utilized a multilayer perceptron (MLP) to represent scenes as continuous volumetric functions. While NeRF excelled at view synthesis, it required multi-view supervision. The leap to text-to-3D was facilitated by DreamFusion [2], which introduced Score Distillation Sampling (SDS). SDS utilizes a pretrained 2D diffusion model  $\epsilon_\phi$  to provide gradients for a 3D model  $\theta$  by minimizing the difference between added and predicted noise.

$$\nabla_\theta \mathcal{L}_{\text{SDS}} = \mathbb{E}_{t,\epsilon} \left[ w(t) (\hat{\epsilon}_\phi(x_t; y, t) - \epsilon) \frac{\partial x}{\partial \theta} \right]$$

Despite enabling high-fidelity synthesis, SDS-based methods frequently suffer from "geometric hallucinations" structural errors where 2D models lack the 3D awareness to maintain global consistency.

#### 3.2 Defining 3D Hallucinations and the Janus Problem

In 3D generative literature, "hallucination" refers to the emergence of geometrically inconsistent or semantically redundant features that do not exist in the source prompt [8]. The most pervasive hallucination is the Janus Problem, characterized by the duplication of canonical features (e.g., multiple faces or sets of limbs) [2], [5], [6]. This occurs because 2D diffusion models are biased toward high-probability frontal views in their training data. As noted by Wang et al.[5], this is a manifestation of "mode collapse," where the optimization settles into a local minimum of a specific 2D viewpoint. Additionally, texture hallucinations often manifest as over-saturation or "color bleeding," a byproduct of the high guidance scales required for convergence in models like TextMesh [9] and Latent-NeRF [10].

#### 3.3 Evolution of Structural Fidelity and Disentanglement

To improve geometric accuracy, researchers moved beyond simple volume density. Magic3D [3] utilized a coarse-to-fine strategy with hash grids to achieve higher resolution, while Fantasia3D [4] introduced a disentangled framework using Physically Based Rendering (PBR) and surface normals to refine geometry separately from appearance. Furthermore, recent breakthroughs in 3D Gaussian Splatting (3DGS) [11] have offered faster rendering speeds, yet these explicit representations remain vulnerable to the same 2D prior induced hallucinations. Variational Score Distillation (VSD), introduced in ProlificDreamer [12], attempted to solve consistency by modeling the 3D scene as a distribution rather than a single point, though it increased computational complexity significantly.

#### 3.4 Mathematical Mitigation and Score-Space Debiasing

Early attempts to detect and mitigate these hallucinations focused on mathematical adjustments to the score space. Perp-Neg [6] proposed a perpendicular gradient sampling method, ensuring that negative prompts (e.g., "side view") do not interfere with the primary semantics of the frontal prompt. Similarly, Entropic Score Distillation (ESD) [5] introduced an entropy regularization term to the objective function -

$$J_{\text{Ent}} = J_{\text{MLE}} - \lambda \mathbb{E}[\Omega(t)H[q_t^\theta(x_t | y)]]$$

By maximizing entropy, ESD [5] encourages view diversity and reduces mode-seeking behavior. However, these methods are "geometry blind" they adjust the gradients without explicitly "understanding" if the generated structure is anatomically or physically correct.

### 3.5 The Multi-Modal Gap - Analyzing Hallo3D and its Limitations

The most recent shift involves the use of Large Multi-modal Models (LMMs) to simulate human-like reasoning. Hallo3D [7] introduced a "generation-detection-correction" paradigm, leveraging models like GPT-4V to identify structural hallucinations (e.g., "a bird with two beaks") and generating "enhanced negative prompts" to guide the diffusion process.

However, critical analysis of [7] reveals a significant open challenge - Multi-modal Reasoning Hallucinations. Current LMMs are not natively 3D-aware and often provide inaccurate feedback or fail to perceive subtle geometric errors that cannot be easily verbalized. Furthermore, Hallo3D relies on a semantic bottleneck; it converts a visual error into text, which is then re-interpreted by a 2D diffusion model. This often leads to "over-correction" or the introduction of new artifacts that the LMM cannot track in real-time.

### 3.6 Rationale for Proposed Research

Existing research has moved from pure 2D priors [2] to mathematical debiasing [5], [6] and text-based multi-modal detection [7]. Yet, the field lacks a closed-loop system that grounds LMM reasoning into spatially aware geometric constraints. This proposed research aims to overcome the limitations of Hallo3D [7] by replacing the semantic bottleneck with a direct visual geometric feedback mechanism. By integrating LMM detection with explicit 3D structural priors, this project will address the "secondary hallucinations" found in multi-modal pipelines, positioning the contribution as a definitive solution for consistent, structurally sound 3D content generation.

## 4 Problem Definition / Research Gap

To address the "semantic bottleneck" and reasoning hallucinations in current text-based 3D generation loops (e.g., Hallo3D [7], Magic3D [3]), this research proposes replacing imprecise "negative prompts" with **direct geometric grounding**. This feedback mechanism aims to provide the spatial precision required to eliminate structural inconsistencies and ensure topological integrity.

## 5 Proposed Research Outcome

### 5.1 System Overview

This research proposes an Adversarial Geometric Distillation (AGD) framework as a refinement layer for text-to-3D generation. Unlike Hallo3D [7], which uses multi-modal models for detection via textual inconsistencies, AGD employs an auxiliary adversarial critic (GNN-based) to identify hallucinations through topological metrics (e.g., Laplacian variance for blobs, Betti numbers for genus), then grounds refinement via self-distillation, converting detections into spatial energy functions (e.g.,  $E(\mathbf{x}) = \lambda L + (1 - \lambda)D$ , where  $L$  is geometry loss and  $D$  is distillation term) for structurally accurate optimization.

## 5.2 Methodology and Architecture

The AGD framework integrates an Auxiliary AI Critic into the 3D refinement loop to ensure topological integrity.

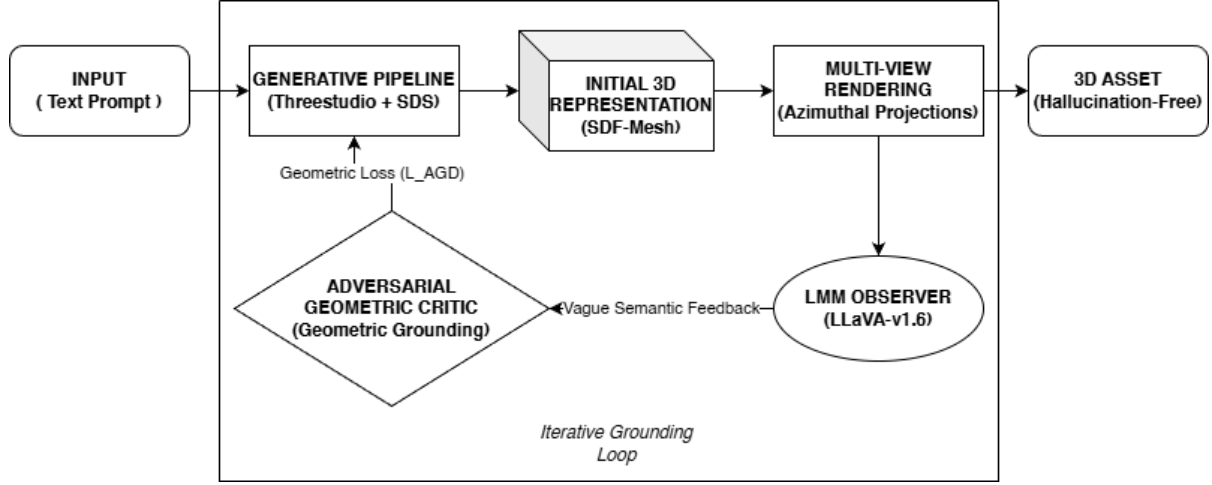


Figure 1: System Architecture of Adversarial Geometric Distillation

## 5.3 Methodology and Architecture

- **Generative Pipeline** - SDS-based synthesis of initial SDF-mesh from text/single-image input using 2D diffusion priors.
- **Multi-View Renderer** - Produces diverse 2D projections for spatial analysis.
- **LMM Hallucination Detector** - Large multimodal model scans renderings for structural issues (e.g., Janus faces, view inconsistencies).
- **Adversarial Geometric Critic** - Translates LMM feedback into a differentiable 3D energy function that localizes anomalies in spatial coordinates/regions.
- **Grounded Refinement Loop** - Iterative optimization adds geometric energy as adversarial loss to directly refine 3D parameters (SDF/mesh), enforcing manifold consistency without text re-prompting.

## 5.4 Process Flow

- **Input** - A text prompt triggers the base 3D synthesis.
- **Detection** - The LMM detects hallucinations.
- **Mapping** - Critic maps detections to 3D energy function (localizes issues).
- **Refinement** - Optimizer refines geometry via combined SDS + geometric loss until artifacts vanishes.
- **Output** - Clean, consistent, photorealistic 3D asset.

## 5.5 Feasibility and Resource Capability

Built in Python with Threestudio + PyTorch. Leverages exist hallucination taxonomy. GPU-friendly (modest overhead from LMM + iterations). Completable in proposed timeline (see Appendix).

## 6 Resource Requirements

The successful implementation of the Adversarial Geometric Distillation (AGD) framework requires the following technical and non-technical resources:

## 6.1 Hardware Resources

High-performance workstation equipped with an NVIDIA RTX 3090/4090 or A100 GPU (minimum 24GB VRAM) to support simultaneous latent diffusion processing and Large Multi-modal Model (LMM) inference.

## 6.2 Software Resources

Frameworks & Libraries: PyTorch for deep learning, Threestudio for 3D generation, and HuggingFace Diffusers for diffusion priorities.

Models & APIs: Local deployment of LLaVA-v1.6 for hallucination detection.

Datasets: Objaverse and Google Scanned Objects (GSO) for geometric benchmarking and evaluation.

## 7 Summary

This research introduces Adversarial Geometric Distillation (AGD) to mitigate hallucinations in text-to-3D generation, focusing on geometric artifacts like genus inconsistencies and blob problems caused by mode collapse in score distillation frameworks. By integrating an auxiliary adversarial critic (GNN-based for topology detection) with grounded self-distillation (teacher-student refinement using Laplacian metrics), AGD replaces view-biased 2D priors with explicit spatial constraints, yielding consistent, manifold-sound 3D meshes without relying on imprecise negative prompts or external multi-modal reasoning. The outcome is a novel, efficient pipeline that reduces Janus artifacts and enhances topological accuracy, as validated on datasets like Objaverse.

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## 9 Appendix – Plan of Action

